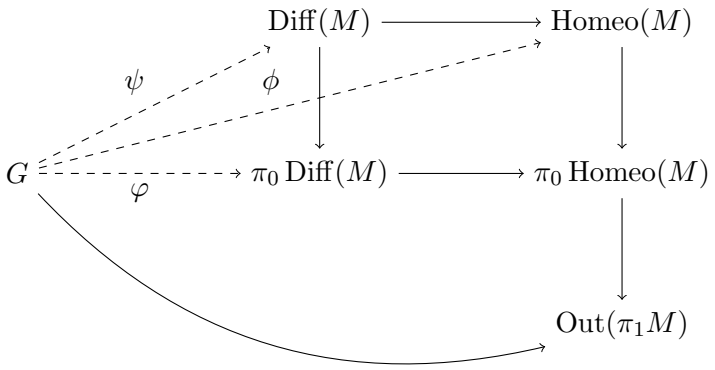
ASYMMETRY AND EXOTIC TORI

OLIVER H. WANG

ABSTRACT. Let p be an odd prime and let Σ^n be a homotopy n-sphere. We show that there are infinitely many exotic tori of the form $M = T^n \# \Sigma^n$ such that there is an order p-element in $\pi_0 \operatorname{Diff}(M)$ which does not lift to an order p-element in $\operatorname{Diff}(M)$. As a consequence, these exotic tori do not admit smooth $\mathrm{SL}_n(\mathbb{Z})$-actions even though the $\mathrm{SL}_n(\mathbb{Z})$-action on homology lifts to $\pi_0 \operatorname{Diff}(M)$. As an application of our techniques, we show that, for every odd prime p, there is a $(2p^2 - 2p - 2)$-dimensional exotic torus of the form above with no smooth, nontrivial $\mathrm{SL}_{2p-2}(\mathbb{Z})$-action. Unlike previous work on smooth Nielsen realization, our methods rely on computations involving the stable homotopy groups of spheres.

1. INTRODUCTION

The term "Nielsen realization problem" encompasses many different types of lifting problems occurring in automorphism groups of manifolds. For a smooth manifold M, one may ask whether or not a subgroup of $\pi_0 \operatorname{Diff}(M)$ lifts to an isomorphic subgroup of $\operatorname{Diff}(M)$. If M is aspherical, it is also interesting to ask whether or not a subgroup of $\operatorname{Out}(\pi_1 M)$ lifts to an isomorphic subgroup of $\operatorname{Diff}(M)$, $\pi_0 \operatorname{Diff}(M)$, $\operatorname{Homeo}(M)$ or $\pi_0 \operatorname{Homeo}(M)$. These questions may be organized by the following diagram.



When M has additional structure, one may also ask for diffeomorphisms that preserve this structure. In this paper, we focus only on lifting to the diffeomorphism group.

Work on Nielsen realization has taken place predominantly in lower dimensions. Kerckhoff showed, for instance that every finite subgroup of the mapping class group of a hyperbolic surface may be realized by isometries [Ker83].

In higher dimensions, some results are known. Schultz [Sch79] studied the case where $M = S^n$ and $n \geq 5$. In this case, $\pi_0 \operatorname{Diff}(S^n)$ is the group of homotopy $(n+1)$-spheres Θ_{n+1}, which is finite and Schultz showed that there are both elements which an be realized as diffeomorphisms as well as elements which cannot.

There are many known examples of high dimensional smooth aspherical manifolds and subgroups $G \leq \operatorname{Out}(\pi_1 M)$ for which the lift ψ does not exist. Raymond–Scott [RS77] find examples of aspherical manifolds M and finite subgroups $G \leq \pi_0 \operatorname{Diff}(M)$ such that a lift ϕ of the diagram above does not exist. In [BKKT24], the authors precisely determine the exotic spheres Σ^n, for which the map $\pi_0 \operatorname{Diff}^+(T^n \# \Sigma^n) \to \mathrm{SL}_n(\mathbb{Z})$ splits. They show that the splitting fails only for certain even order elements of Θ_n (of which there are infinitely many as n varies). In particular, when Σ^n has odd order in Θ_n, one may still ask whether $\mathrm{SL}_n(\mathbb{Z})$ lifts to $\operatorname{Diff}(T^n \# \Sigma^n)$.

Our focus in this paper will address the passage from $\pi_0 \operatorname{Diff}(M)$ to $\operatorname{Diff}(M)$. By [BKKT24, Theorem D], every p-torsion element of $\mathrm{SL}_n(\mathbb{Z})$ lifts to $\pi_0 \operatorname{Diff}(M)$. In fact, we may choose this lift so that a lift $\phi : C_p \to \operatorname{Homeo}(M)$ exists. We show that for many such Σ^n, there are finite cyclic groups in $\mathrm{SL}_n(\mathbb{Z})$ which fail to lift to the diffeomorphism group. We prove the following.

Theorem 1.1. *There are infinitely many exotic tori of the form $\mathcal{T}^n = T^n \# \Sigma^n$ such that there is a faithful homomorphism $\varphi : C_p \to \mathrm{SL}_n(\mathbb{Z})$ which does not lift to $\operatorname{Diff}(\mathcal{T}^n)$.*

As a corollary, we obtain the following.

Corollary 1.2. *For $\mathcal{T}^n$ as in Theorem 1.1, there is a faithful homomorphism $\varphi : C_p \to \pi_0 \operatorname{Diff}(M)$ which does not lift to $\operatorname{Diff}(M)$.*

Applying [BKKT24, Theorem B], we obtain examples of exotic tori without $\mathrm{SL}_n(\mathbb{Z})$-actions.

Corollary 1.3. *There is no smooth, nontrivial $\mathrm{SL}_n(\mathbb{Z})$-action on the exotic tori of Theorem 1.1.*

The primary conceptual tool in proving Theorem 1.1 is equivariant smoothing which was studied in [Las79, LR78]. In order to obtain explicit examples, however, we must appeal to computations involving the stable homotopy groups of spheres. When Σ^n is an exotic sphere arising from $\beta_1 \in \pi^s_{2p^2-2p-2}$, we obtain a particularly interesting family of examples.

Theorem 1.4. *For every odd prime p, there are $(2p^2-2p-2)$-dimensional exotic tori such that every nontrivial smooth C_p-action has a (p^2-p-2)-dimensional fixed*

set. Additionally, these exotic tori do not admit any smooth, nontrivial $\mathrm{SL}_{2p-2}(\mathbb{Z})$*-actions.*

1.1. **Acknowledgments.** The author was supported by the NSF under Award No. DMS-1839968.

2. Classifying spaces for proper actions

Definition 2.1. Let Γ be a discrete group. A *proper action* of Γ on a space X is one such that the point stabilizers are finite.

Definition 2.2. Let Γ be a discrete group. We let $\underline{E}\Gamma$ denote the Γ-space such that $(\underline{E}\Gamma)^H \simeq \begin{cases} * & H \text{ finite} \\ \emptyset & H \text{ infinite} \end{cases}$. We say that $\underline{E}\Gamma$ is the *classifying space for proper actions.*

The space $\underline{E}\Gamma$ is a classifying space in the following sense. If X is a Γ-space with a proper action, then there is a Γ-equivariant map $X \to \underline{E}\Gamma$ which is unique up to Γ-equivariant homotopy. Classifying spaces for proper actions can always be constructed but many exist in nature. The main example that will be relevant to us is the following.

Example 1. Let M be a connected complete non-positively curved manifold and let C_p act on M by isometries and such that there is a fixed point. Let $\Gamma = \pi_1 M \rtimes C_p$ where the semi-direct product is constructed with the evident action of C_p on $\pi_1 M$. Then, Γ acts on the universal cover $\tilde{M}$ and $\tilde{M} \simeq \underline{E}\Gamma$.

Lemma 2.3. *Suppose C_p acts locally linearly and with a fixed point on a closed aspherical manifold M with fundamental group π. Let $\Gamma = \pi \rtimes C_p$. Suppose there is a model of $\underline{E}\Gamma$ such that the quotient $\underline{E}\Gamma/\pi$ is a finite CW-complex. Let $\tilde{f} : \tilde{M} \to \underline{E}\Gamma$ be the universal Γ-equivariant map and let $f : M \to \underline{E}\Gamma/\pi$ denote the C_p-equivariant map on quotients. Then f is a homotopy equivalence and $f|_{M^{C_p}} : M^{C_p} \to (\underline{E}\Gamma/\pi)^{C_p}$ is a mod-p homology equivalence.*

Proof. As a non-equivariant map, f is a homotopy equivalence since $\tilde{f}$ is a π-equivariant map of contractible free π-spaces. The cofiber C of f is a finite C_p-CW-complex which is non-equivariantly contractible. It follows from Smith theory that C^{C_p} is mod-p acyclic. Since the fixed set of the cofiber is the cofiber of the map of fixed sets, f restricts to a mod-p homology equivalence on fixed sets. □

Specializing to the case of tori, a (not necessarily free) C_p-action on T^n induces an action on $H_1(T^n)$ yielding a group homomorphism $\rho : C_p \to \mathrm{SL}_n \mathbb{Z}$. Any group homomorphism $\rho : C_p \to \mathrm{SL}_n \mathbb{Z}$ determines an action of C_p on T^n by isometries with a fixed point, which we call the linear action. Let T^n_ρ denote the torus with the linear action determined by ρ. Note that the universal cover of T^n_ρ is a model for

$\underline{E}\Gamma$, where Γ is defined as above. Applying Lemma 2.3, we may compare smooth C_p-actions on exotic tori with the linear action on the standard torus.

Corollary 2.4. *Let C_p act on T^n locally linearly such that there is a fixed point and let $\rho : C_p \to \mathrm{SL}_n \mathbb{Z}$ be given by the action on π_1. Then the C_p-equivariant map $f : T^n \to T^n_\rho$ is a non-equivariant homotopy equivalence and restricts to a mod-p homology equivalence on the fixed set.*

3. Equivariant smoothing and actions on exotic spheres

Smoothing theory was developed in [KS77] and the equivariant versions were studied in [LR78, Las79]. We briefly summarize the relevant material here.

A *smooth structure* or a *smoothing* of a topological manifold N is represented by a homeomorphisms $\eta : M \to N$ where M is a smooth manifold. We say that two smooth structures $\eta_i : M_i \to N$, $i = 0, 1$, are *isotopic* if there is a diffeomorphism $\alpha : M_0 \to M_1$ and an isotopy $H : M_0 \times I \to N$ restricting to η_0 on $M_0 \times \{0\}$ and $\eta_1 \circ \alpha$ on $M_0 \times \{1\}$. Let $\mathcal{S}(N)$ denote the pointed set of isotopy classes of smooth structures of N and, if $\eta \in \mathcal{S}(N)$, let $\mathcal{S}(N, \eta)$ denote the pointed set of isotopy classes of smooth structures with basepoint η. The main result of [KS77, Essay IV] is that isotopy classes of smooth structures on N are in bijection with linear reductions of the stable tangent microbundle provided $\dim N \geq 5$. If N has at least one smooth structure, this yields a bijection of pointed sets $\delta_\eta : \mathcal{S}(N, \eta) \xrightarrow{\cong} [N, TOP/O]$ where TOP/O is an infinite loop space.

Since TOP/O is an infinite loop space, $[N, TOP/O]$ is always an abelian group. When $N = S^n$ and $n \geq 5$, this abelian group is Kervaire–Milnor's group of homotopy spheres Θ_n. The elements are oriented homotopy spheres and the group operation is given by connect sum. One of the main results of [KM63] is that Θ_n is finite.

It is sometimes the case that two smooth structures $f_i : M_i \to N$, $i = 0, 1$, are distinct even though M_0 and M_1 are diffeomorphic. For example, if Σ^n is an exotic n-sphere, the evident are homeomorphisms $(T^n \# \Sigma^n) \times T^m \to T^{n+m}$ and $T^m \times (T^n \# \Sigma^n) \to T^{n+m}$ are not isotopic but the domains are diffeomorphic. The set $\mathcal{S}(N, \eta)/\operatorname{Homeo}(N)$ classifies smooth manifolds homeomorphic to N up to diffeomorphism (note that the action of $\operatorname{Homeo}(N)$ on the set of smooth structures is not necessarily basepoint preserving).

Suppose G is a compact Lie group. For manifolds N with locally linear G-actions, one can make analogous definitions and attempt to carry out Kirby–Siebenmann's work in the equivariant setting. Equivariant smoothing was developed by Lashof–Rothenberg in [LR78] but, due to the lack of an equivariant product structure theorem, the theory is much more difficult to work with. One may instead study stable equivariant smoothings which are represented by an equivariant homeomorphism $F : M \to N \times V$ for some G-representation V. We refer to [Las79] for the definition of stable isotopy of these representatives. For a smooth G-manifold N, let $\mathcal{S}^{st}_G(N, \eta)$

denote the stable equivariant smoothings of N. The space TOP/O can be given the structure of an equivariant infinite loop space and there is a bijection of pointed sets $\mathcal{S}^{st}_G(N,\eta) \cong [N, TOP/O]^G$ when N has finitely many orbit types [Las79].

When N has a locally linear G-action, there there is a restriction map

$$\mathrm{res}_N : [N, TOP/O]^{C_p} \to [N, TOP/O]$$

which forgets equivariance. In terms of smoothings, this sends a stable C_p-smoothing of N to the underlying non-equivariant smoothing of $N \times \mathbb{R}^i$. The product structure theorem [KS77, Essay I] states that isotopy classes of smoothings of $N \times \mathbb{R}^i$ are in bijection with those of N, so we obtain a smoothing on N. When working with C_p-actions, it is often useful to localize TOP/O at the prime p. This can be done equivariantly [MMT82] and we will use $(\mathrm{res}_N)_{(p)}$ to denote the evident restriction map.

We will primarily be interested in C_p-actions on exotic tori of the form $\mathcal{T}^n = T^n \# \Sigma^n$. In the case $\Sigma^n = p\Sigma'$ for some $\Sigma' \in \Theta_n$, any smooth C_p-action on T^n gives a smooth C_p-action on $\mathcal{T}^n$ by taking an equivariant connect sum. So, we will primarily be interested in C_p-actions when Σ^n is not divisible by p.

We record some previously known facts about homotopy tori.

Proposition 3.1. *Suppose $\mathcal{T}^n = T^n \# \Sigma^n \to T^n$ is a smoothing of the n-torus. Then, the following hold:*

(1) $[\psi : \mathcal{T}^n \to T^n]$ is invariant under the action of $\mathrm{Homeo}(T^n)$ on $\mathcal{S}(T^n)$.
(2) If Σ^n is not divisible by p in Θ_n, then $\mathcal{T}^n$ does not have a free C_p-action.

The first is contained in the proof of [BKKT24, Lemma 3.1] and the second is [BT25, Theorem B]. The first part of Proposition 3.1 tells us that $[\psi_0 : \mathcal{T}^n \to T^n] = [\psi_1 : \mathcal{T}^n \to T^n]$ for any homeomorphisms ψ_0 and ψ_1. Therefore, we may unambiguously speak of $[\mathcal{T}^n] \in \mathcal{S}(T^n)$. After identifying $\mathcal{S}(T^n)$ with $[T^n, TOP/O]$ using the standard smooth structure, $[\mathcal{T}^n]$ determines an element of $[T^n, (TOP/O)_{(p)}]$, which we also denote $[\mathcal{T}^n]$.

Proposition 3.2. *Suppose $\mathcal{T}^n = T^n \# \Sigma^n$ where Σ^n is not divisible by p in Θ_n. Suppose $\mathcal{T}$ has a smooth action of C_p let ρ denote the action of C_p on $H_1(\mathcal{T}^n)$. Then $[\mathcal{T}^n]$ is in the image of the localized restriction map*

$$(\mathrm{res}_{T^n_\rho})_{(p)} : [T^n_\rho, (TOP/O)_{(p)}]^{C_p} \to [T^n, (TOP/O)_{(p)}].$$

Proof. Since Σ^n is not divisible by p, the C_p-action must have a fixed point so we may apply Corollary 2.4 to obtain a C_p-equivariant p-local homology equivalence $f : \mathcal{T}^n \to T^n$.

The result follows from applying [KQW, Lemma 3.1] but, for the reader's convenience, we outline the main idea. By [MMT82, Theorem 9], f induces isomorphisms

in the diagram

$$\begin{array}{ccc} [T^n_\rho, (TOP/O)_{(p)}]^{C_p} & \xrightarrow{\cong} & [\mathcal{T}^n, (TOP/O)_{(p)}]^{C_p} \\ \Big\downarrow \text{res} & & \Big\downarrow \text{res} \\ [T^n, (TOP/O)_{(p)}] & \xrightarrow{\cong} & [\mathcal{T}^n, (TOP/O)_{(p)}] \end{array}.$$

We would like to interpret the top map as a map between sets of stable C_p-smoothings and we would like to interpret the bottom map as a map between sets of smoothings. Then the result follows from tracking the stable C_p-smoothing given by the identity on $\mathcal{T}^n$ through the diagram. The main technical issue here is that, in order to identify the objects in the diagram above with sets of smoothings, we must translate them into lifting problems and the tangent C_p-microbundle of $\mathcal{T}^n$ need not be compatible with the tangent C_p-microbundle of T^n_ρ. To resolve this issue, we replace $\mathcal{T}^n$ with the total space of a certain C_p-vector bundle over $\mathcal{T}^n$. Details may be found in [KQW, Lemma 3.1].

As every homotopy equivalence of a torus is homotopic to a homeomorphism and exotic tori of the form $T^n \# \Sigma^n$ are stably parallelizable, [KQW, Lemma 3.1] implies that $[\psi : \mathcal{T}^n \to T^n] \in \mathcal{S}(T^n)$ is in the image of the restriction for some homeomorphism ψ. □

3.1. **Actions on homotopy spheres.** Recall that, if p is an odd prime and $0 \leq k \leq \frac{n}{2}$, then there is a smooth C_p-action on the standard sphere S^n with fixed set S^{n-2k}. In [KQW], it is shown that if a homotopy sphere Σ has a C_p-action with isotropy representation V, then Σ is in the image of the localized restriction map $(\text{res}_{S^V})_{(p)}$. Computations in stable homotopy theory show that some homotopy spheres are not in the image of certain restriction maps. Smooth C_p-actions on these homotopy spheres are more restricted in the sense that there may be values of k such that there is no smooth C_p action with codimension $2k$ fixed set.

We give some explicit examples of this behavior. Recall the Kervaire–Milnor sequence

$$0 \to bP_{n+1} \to \Theta_n \xrightarrow{PT} \operatorname{coker} J_n.$$

Here, bP_{n+1} is the subgroup which bounds parallelizable $(n+1)$-manifolds and J_n is the J-homomorphism. The sequence is exact and the cokernel of the right map is either 0 or $\mathbb{Z}/2$. The map PT can be described by a Pontryagin–Thom construction. Let π^s_* denotes the stable homotopy groups of spheres. For an element $x \in \operatorname{coker} J_n$, let $\widetilde{x} \in \pi^s_n$ denote a lift. For a fixed p, there is an element $\alpha_1 \in \pi^s_{2p-3}$ of order p. The following result is in [KQW] (an equivalent version is given as [Sch75, Theorem 4.18]).

Theorem 3.3 (Schultz 1975, Kuhn–Quigley–Wang 2026)**.** *Suppose Σ is a homotopy sphere such that $\alpha_1\widetilde{PT(\Sigma)}$ is not divisible by p. Then $PT(\Sigma)$ is not in the image of res_{S^V} unless $|V|-|V^{C_p}|$ is a multiple of $2p$. In particular, any C_p action on Σ must have a fixed set whose codimension is a multiple of $2p$.*

Example 2. In the case $p = 3$, $\alpha_1 \in \pi_3^s$. There is an element $\beta_1 \in \pi_{10}^s$ such that $\alpha_1\beta_1 \in \pi_{13}^3$ is not divisible by 3. It follows that, if a homotopy sphere Σ has Pontryagin–Thom invariant β_1, then any smooth C_3-action on Σ must have 4-dimensional fixed set.

This is an instance of a more general phenomenon. For odd primes p, there is an element $\beta_1 \in \pi^s_{2(p^2-p-1)}$ such that $\alpha_1\beta_1$ is not divisible by p.

Example 3. There are also examples of exotic spheres with restricted C_p-actions which do not fit in the framework of Theorem 3.3. For $p = 3$, it turns out that $\alpha_1(\alpha_1\beta_1)$ is divisible by 3 so the theorem does not apply to $\alpha_1\beta_1$. However, any smooth, non-free C_3-action on such a homotopy 13-sphere must have fixed set of codimension $6, 8$ or 12.

We make the following definition.

Definition 3.4. For a prime p and positive integers m, n, define $\mathcal{A}_{p,n,m} \subseteq \Theta_n$ to be those homotopy n-spheres not in the image of $(\mathrm{res}_{S^V})_{(p)}$ for any C_p-representation V with $|V^{C_p}| = m$.

4. Failure of Nielsen realization

To show that Nielsen realization fails for many exotic tori, we show that the fixed point restrictions of C_p-actions on exotic spheres give analogous fixed point restrictions of C_p-actions on $T^n\#\Sigma$.

Proposition 4.1. *Let $\Sigma \in \mathcal{A}_{p,n,m}$ and let $\rho : C_p \to \mathrm{GL}_n(\mathbb{Z})$ be a representation such that T^n_ρ has a fixed set of dimension m. Then $\mathcal{T}^n := T^n\#\Sigma$ does not have a smooth C_p-action with induced action ρ on $H_1(M)$.*

Proof. Throughout the proof, we implicitly localize TOP/O at p.

By Proposition 3.1, there is no free smooth C_p-action. By Proposition 3.2, it suffices to show that M is not in the image of $(\mathrm{res}_{T^n_\rho})_{(p)}$.

There is a spectral sequence

$$E^{i,-j}_{2,C_p}(T^n_\rho) = H^i_{C_p}(T^n_\rho; \underline{\pi}_j TOP/O) \Rightarrow [\Sigma^{i+j}T^n_\rho, TOP/O]^{C_p}$$

where the left hand side is Bredon cohomology with coefficients in the C_p-Mackey functor $\underline{\pi}_j TOP/O$. By forgetting the group action, there is a map from this spectral sequence to the Atiyah–Hirzebruch spectral sequence

$$E_2^{i,-j}(T^n) = H^i(T^n; \pi_j TOP/O) \Rightarrow [\Sigma^{i+j}T^n, TOP/O]$$

for the cohomology theory determined by TOP/O.

Let V be the isotropy representation at a fixed point of T^n_ρ. Then the equivariant collapse map $T^n_\rho \to S^V$ gives the following commuting diagram.

$$\begin{array}{ccc} H^n_{C_p}(S^V;\underline{\pi}_n TOP/O) & \xrightarrow{\cong} & H^n_{C_p}(T^n_\rho;\underline{\pi}_n TOP/O) \\ \downarrow & & \downarrow \\ E^{n,-n}_{\infty,C_p}(S^V) & \longrightarrow & E^{n,-n}_{\infty,C_p}(T^n_\rho) \\ \downarrow & & \downarrow \\ E^{n,-n}_{\infty}(S^{|V|}) & \xrightarrow{\cong} & E^{n,-n}_{\infty}(T^n) \\ \cong\downarrow & & \downarrow \\ [S^{|V|},TOP/O] & \longrightarrow & [T^n,TOP/O] \end{array} \tag{1}$$

The upper vertical maps are surjections for degree reasons. Since C_p acts freely on the n-cells and $(n-1)$-cells of S^V and T^n, the top map may be identified with

$$H^n(S^V/C_p;\pi_n TOP/O) \to H^n(T^n/C_p;\pi_n TOP/O).$$

Taking cap products with a fundamental class $[S^V/C_p] \in H_n(S^V/C_p) \cong \mathbb{Z}$ yields an isomorphism

$$H^n(S^V/C_p, S^{V^{C_p}};\pi_n TOP/O) \to H_0((S^V/C_p)\setminus S^{V^{C_p}};\pi_n TOP/O)$$

by Lefschetz duality. It follows from the diagram

$$\begin{array}{ccc} H^n(S^V/C_p) & \xrightarrow{-\cap[S^V/C_p]} & H_0(S^V/C_p) \\ \cong\uparrow & & \cong\uparrow \\ H^n(S^V/C_p, S^{V^{C_p}}) & \xrightarrow{-\cap[S^V/C_p]} & H_0((S^V/C_p)\setminus S^{V^{C_p}}) \end{array}$$

that $-\cap[S^V/C_p] : H^n(S^V/C_p;\pi_n TOP/O) \to H_0(S^V/C_p;\pi_n TOP/O)$ is an isomorphism. Similarly, taking cap products with the fundamental class of T^n/C_p gives an isomorphism $H^n(T^n/C_p;\pi_n TOP/O) \to H_0(T^n/C_p;\pi_n TOP/O)$. One can check that the map $T^n/C_p \to S^V/C_p$ preserves fundamental classes. It follows from properties of the cap product that the top map is an isomorphism.

Stably, tori are wedges of spheres so

$$H^n(T^n;\pi_n TOP/O) = E_2^{n,-n}(T^n) = E_\infty^{n,-n}(T^n)$$

and this is a summand of $[T^n, TOP/O]$. Moreover, since we have localized at an odd prime, the group $\mathrm{Homeo}(T^n)$ acts trivially on this summand. Finally, the bottom map of Diagram 1 sends a homotopy sphere Σ to $T^n\#\Sigma$. So if Σ is not in the image of res_{S^V} then $T^n\#\Sigma$ is not in the image of res_{T^n}. □

Proof of Theorem 1.1. To finish the proof of Theorem 1.1, it suffices to show that, for a fixed p, there are infinitely many actions ρ and homotopy spheres Σ satisfying the hypotheses of Proposition 4.1. As there are infinitely many Σ satisfying the hypotheses of Theorem 3.3), there are infinitely many Σ in $A_{p,n,n-(p-1)}$ (as n varies). Let $\rho : C_p \to \mathrm{SL}_n(\mathbb{Z})$ be the sum of the reduced regular representation and $(n-p+1)$-many copies of the trivial representation. The linear C_p-action on T^{p-1} given by the reduced regular representation has a discrete fixed set [DL21] so the linear C_p-action on T^n given ρ has $(n-p+1)$-dimensional fixed set. By Proposition **??**, the homotopy torus $M = T^n\#\Sigma$ does not have a free C_p-action. It follows from Proposition 4.1, $T^n\#\Sigma$ does not have a smooth C_p-action inducing ρ on the first homology. □

Remark. In fact, the fixed set of any linear C_p-action on T^n must have codimension a multiple of $p-1$ and all such codimensions are realized by some action. This follows from the fact that a finitely generated irreducible $\mathbb{Z}[C_p]$-module is rationally $\mathbb{Q}$ with a trivial action, the regular representation $\mathbb{Q}[C_p]$ or the reduced regular representation $\mathbb{Q}[C_p]/\mathbb{Q}$. There are $\left|\tilde{K}_0(\mathbb{Z}[C_p])\right|$-many irreducible modules which rationalize to $\mathbb{Q}[C_p]$ and there are $\left|\tilde{K}_0(\mathbb{Z}[C_p])\right|$-many irreducible modules which rationalize to $\mathbb{Q}[C_p]/\mathbb{Q}$. For homotopy n-spheres Σ of Theorem 3.3, one may use these facts to construct many examples of p-torsion elements of $\mathrm{GL}_n(\mathbb{Z})$ do not lift to $\mathrm{Diff}(T^n\#\Sigma)$.

Remark. The main theorem of [KQW] implies a partial converse of Proposition 4.1. If $\Sigma \in \Theta_n$ is not in $\mathcal{A}_{p,n,m}$ and $2 < m < \frac{n}{2}$ then Σ has a smooth C_p-action such that it is equivariantly homeomorphic to a representation sphere. Moreover, if V is an n-dimensional C_p-representation with m-dimensional fixed set, then the isotropy representation can be taken to be V. Thus, given a smooth C_p-action on T^n with m-dimensional fixed set, we may take an equivariant connect sum to obtain a smooth C_p-action on $T^n\#\Sigma$ and the resulting action will be topologically conjugate to the original.

5. Actions of $\mathrm{SL}_{2p-2}(\mathbb{Z})$

In this section, we fix an odd prime p and let $n = 2p^2 - 2p - 2$. Let Σ^n denote an exotic n-dimensional sphere with Pontryagin–Thom invariant β_1. Any smooth C_p-action on Σ^n must have codimension a multiple of $2p$. Smooth actions on the exotic torus $T^n\#\Sigma^n$ are further restricted. As any smooth, non-free C_p-action on a torus must have fixed set with codimension a multiple of $p-1$, it follows that any such action on $T^n\#\Sigma^n$ must have fixed set with codimension a multiple of $p(p-1)$.

For our choice of n, this forces the fixed set of a nontrivial smooth C_p-action to be (p^2-p-2)-dimensional.

In $\mathrm{SL}_{2p-2}(\mathbb{Z})$, there are two distinguished conjugacy classes of order p-subgroups that we will be concerned with. Let $\tilde{\rho}$ denote the reduced regular representation. Let G denote the order p-subgroup corresponding to the representation $\tilde{\rho} \oplus \mathbb{Z}^{p-1}$ and let H denote the order p-subgroup corresponding to the representation $\tilde{\rho} \oplus \tilde{\rho}$.

Lemma 5.1. *Suppose $f : \mathrm{SL}_{2p-2}(\mathbb{Z}) \to \mathrm{SL}_n(\mathbb{Z})$ is a group homomorphism such that the representations $f|_G$ and $f|_H$ have fixed sets of the same rank. Then f is trivial.*

Proof. By Borel density, we obtain the following diagram.

$$\begin{array}{ccc} \mathrm{SL}_{2p-2}(\mathbb{Z}) & \xrightarrow{\quad f \quad} & \mathrm{SL}_n(\mathbb{Z}) \\ \downarrow & & \downarrow \\ \mathrm{SL}_{2p-2}(\mathbb{R}) & \longrightarrow & \mathrm{SL}_n(\mathbb{R}) \end{array}$$

The irreducible $\mathrm{SL}_{2p-2}(\mathbb{R})$-representations of dimension at most n are the trivial representation, the standard representation and the wedge $\wedge^2\mathbb{R}^{2p-2}$ so the bottom map corresponds to a sum of these representations. It is straightforward to check that, unless one only has copies of the trivial representation, the subgroups G and H determine C_p-representations whose fixed sets have different dimensions. So the composition $\mathrm{SL}_{2p-2}(\mathbb{Z}) \to \mathrm{SL}_n(\mathbb{R})$ is trivial. It follows that f must also be trivial. □

We now prove Theorem 1.4.

Proof of Theorem 1.4. Suppose $\mathrm{SL}_{2p-2}(\mathbb{Z}) \to \mathrm{Diff}(M)$ is a homomorphism and consider the composition

$$f : \mathrm{SL}_{2p-2}(\mathbb{Z}) \to \mathrm{Diff}(M) \to \mathrm{SL}_n(\mathbb{Z})$$

where the last map sends a diffeomorphism to the induced map on $H_1(M)$. The fixed set of any nontrivial C_p-action must be (p^2-p-2)-dimensional. In particular, when we consider the representations determined by the actions of H and G on $H_1(M)$, we see that, if both representations are nontrivial, then the fixed set must be of the same rank. Applying Lemma 5.1, we see that either H or G must be in the kernel of f. Note that neither H nor G is contained in a congruence subgroup (as reduced regular representations over $\mathbb{F}_p$ are nontrivial). So, if either H or G is contained in the kernel of a homomorphism, then the homomorphism must be trivial.

It remains to show that the map $\mathrm{SL}_{2p-2}(\mathbb{Z}) \to \mathrm{Diff}(M)$ is trivial provided the homomorphism f is trivial. Since there is no free C_p-action on M, the subgroups

H and G must fix a point. But since the groups must act trivially on homology, it follows that the H and G must be trivial. This implies that $\mathrm{SL}_{2p-2}(\mathbb{Z}) \to \mathrm{Diff}(M)$ is trivial. □

References

[BKKT24] Mauricio Bustamante, Manuel Krannich, Alexander Kupers, and Bena Tshishiku. Mapping class groups of exotic tori and actions by $\mathrm{SL}_d(\mathbf{Z})$. *Trans. Amer. Math. Soc. Ser. B*, 11:1316–1349, 2024.

[BT25] Mauricio Bustamante and Bena Tshishiku. Symmetries of exotic aspherical space forms. *Math. Z.*, 309(3):Paper No. 47, 26, 2025.

[DL21] James F. Davis and Wolfgang Lück. Manifolds homotopy equivalent to certain torus bundles over lens spaces. *Comm. Pure Appl. Math.*, 74(11):2348–2397, 2021.

[Ker83] Steven P. Kerckhoff. The Nielsen realization problem. *Ann. of Math. (2)*, 117(2):235–265, 1983.

[KM63] Michel A. Kervaire and John W. Milnor. Groups of homotopy spheres. I. *Ann. of Math. (2)*, 77:504–537, 1963.

[KQW] Nicholas J. Kuhn, J.D. Quigley, and Oliver H. Wang. Exotic representation spheres. *to appear*.

[KS77] Robion C. Kirby and Laurence C. Siebenmann. *Foundational essays on topological manifolds, smoothings, and triangulations*, volume No. 88 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ; University of Tokyo Press, Tokyo, 1977. With notes by John Milnor and Michael Atiyah.

[Las79] Richard Lashof. Stable G-smoothing. In *Algebraic topology, Waterloo, 1978 (Proc. Conf., Univ. Waterloo, Waterloo, Ont., 1978)*, volume 741 of *Lecture Notes in Math.*, pages 283–306. Springer, Berlin, 1979.

[LR78] R. Lashof and M. Rothenberg. G-smoothing theory. In *Algebraic and geometric topology (Proc. Sympos. Pure Math., Stanford Univ., Stanford, Calif., 1976), Part 1*, volume XXXII of *Proc. Sympos. Pure Math.*, pages 211–266. Amer. Math. Soc., Providence, RI, 1978.

[MMT82] J. P. May, J. McClure, and G. Triantafillou. Equivariant localization. *Bull. London Math. Soc.*, 14(3):223–230, 1982.

[RS77] Frank Raymond and Leonard L. Scott. Failure of Nielsen's theorem in higher dimensions. *Arch. Math. (Basel)*, 29(6):643–654, 1977.

[Sch75] Reinhard Schultz. Differentiable group actions on homotopy spheres. I. Differential structure and the knot invariant. *Invent. Math.*, 31(2):105–128, 1975.

[Sch79] Reinhard Schultz. Isotopy classes of periodic diffeomorphisms on spheres. In *Algebraic topology, Waterloo, 1978 (Proc. Conf., Univ. Waterloo, Waterloo, Ont., 1978)*, volume 741 of *Lecture Notes in Math.*, pages 334–354. Springer, Berlin, 1979.